11 Interval Estimation

A point estimate gives us one number. It does not tell us how much that number might change if we collected another sample. An interval estimate adds this missing information by giving a range with a carefully defined measure of uncertainty.

The main challenge is understanding what that measure means. A confidence interval describes the long-run performance of a sampling procedure. A Bayesian credible interval describes uncertainty in a posterior distribution. Both give ranges, but they answer different probability questions.

This chapter builds confidence intervals from the normal, t, and chi-square distributions we have already studied. We will also return to posterior distributions to see how credible intervals are constructed.

11.1 From a point estimate to a random interval

Imagine estimating a population mean from a sample. Before collecting the observations, the sample mean is random. An interval built around it is random too. A different sample generally gives different endpoints.

> **Notation: interval endpoints and confidence level**
>
> θ denotes the fixed, unknown parameter we want to estimate. The bold symbol $\mathbf{X} = (X_1, \dots, X_n)$ represents the entire random sample.
> $L(\mathbf{X})$ and $U(\mathbf{X})$ are statistics that give the lower and upper endpoints of an interval. We often shorten them to L and U. The expression $[L, U]$ includes all numbers between the endpoints, including the endpoints themselves.
> The Greek letter α is the chosen probability of failing to cover the parameter. The confidence level is $1 - \alpha$. For a 95% confidence level, $\alpha = 0.05$.
> P_θ denotes probability when the population parameter has the particular value θ. Its subscript tells us which sampling distribution is being used.

An exact confidence interval procedure satisfies

$$P_\theta\{L(\mathbf{X}) \leq \theta \leq U(\mathbf{X})\} = 1 - \alpha$$

for every allowed value of θ. The parameter stays fixed in this probability statement. The sample, and therefore the interval, varies.

The probability that the interval contains the parameter is called its **coverage probability**. Some procedures have coverage at least $1-\alpha$, rather than exactly $1-\alpha$. Others have coverage only approximately equal to the stated level. In this chapter, our normal-model confidence intervals have exact coverage under their stated assumptions.

After observing a sample, we substitute the data into the endpoint formulas. The resulting numerical interval is the observed confidence interval.

> **i** Notation: random endpoints and observed endpoints
>
> Lowercase $\mathbf{x} = (x_1, \dots, x_n)$ denotes the observed data. The numbers $l = L(\mathbf{x})$ and $u = U(\mathbf{x})$ are the observed endpoints.
> Uppercase L and U emphasize what varies across possible samples. Lowercase l and u refer to the values calculated from the sample we actually obtained.

11.2 A normal mean when the variance is known

Suppose the observations are independent and each has a normal distribution with mean μ and variance σ^2. For now, assume that σ is known.

> **i** Notation: a mean interval and its normal cutoff
>
> $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is the sample mean.
> μ is the population mean, σ^2 is the population variance, and σ is its positive square root, the population standard deviation. The quantity $\sigma/\sqrt{n}$ is the standard deviation of the sample mean, also called its standard error.
> z_p is the standard normal quantile with probability p to its left. Thus $P(Z \le z_p) = p$ for $Z \sim N(0,1)$.
> The expression $a \pm b$ describes the interval from $a - b$ to $a + b$. The positive quantity b is its margin of error, or half-width.

We know that

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

Leaving probability $\alpha/2$ in each tail gives

$$P\left\{-z_{1-\alpha/2} \le \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \le z_{1-\alpha/2}\right\} = 1-\alpha.$$

To turn this into an interval for μ, multiply through by the positive standard error. Then rearrange both inequalities so that μ is in the middle:

$$P\left\{\overline{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \overline{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1-\alpha.$$

The confidence interval is therefore

$$\boxed{\overline{X} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.}$$

For 95% confidence, the positive cutoff is $z_{0.975} \approx 1.96$. We use 0.975 because the upper tail has probability 0.025. Using $z_{0.95}$ instead would produce a central (equal-tailed) 90% interval.

11.2.1 Worked example: planning for a desired precision

A laboratory will measure the concentration of a reference solution. Assume independent normal measurements with a known standard deviation of 4.5 units. Before collecting data, the laboratory wants to choose a sample size so that its 95% confidence interval for the mean has a margin of error no greater than 1.2 units.

This is a planning question. We do not need a sample mean because the margin of error in the known-variance setting does not depend on the observed measurements.

The requirement is

$$z_{0.975}\frac{4.5}{\sqrt{n}} \le 1.2.$$

Both sides are positive, so we can rearrange and square to obtain

$$n \ge \left(\frac{z_{0.975} \times 4.5}{1.2}\right)^2 \approx 54.021.$$

A sample size must be a whole number. We round upward and plan for **55 measurements**. With 54 measurements, the margin is about 1.2002, just above the target. With 55, it is about 1.1893.

```r
required_size <- (qnorm(0.975) * 4.5 / 1.2)^2
required_size
```

```
[1] 54.02051
```

```r
qnorm(0.975) * 4.5 / sqrt(54)
```

```
[1] 1.200228
```

```r
qnorm(0.975) * 4.5 / sqrt(55)
```

```
[1] 1.189267
```

The formula also shows the cost of greater precision. To cut the margin of error in half while keeping the confidence level and population standard deviation unchanged, we need four times as many observations.

```r
sample_sizes <- 10:150
margin_90 <- qnorm(0.95) * 4.5 / sqrt(sample_sizes)
margin_95 <- qnorm(0.975) * 4.5 / sqrt(sample_sizes)
margin_99 <- qnorm(0.995) * 4.5 / sqrt(sample_sizes)

plot(
  sample_sizes, margin_99,
  type = "l", col = "darkorange", lty = 3, lwd = 2,
  ylim = c(0, 4), xlab = "Number of measurements",
  ylab = "Margin of error"
)
lines(sample_sizes, margin_95, col = "steelblue", lwd = 2)
lines(sample_sizes, margin_90, col = "gray40", lty = 2, lwd = 2)
abline(h = 1.2, col = "gray65", lty = 4)
points(55, qnorm(0.975) * 4.5 / sqrt(55), pch = 16, col = "steelblue")
legend(
  "topright", legend = c("99% confidence", "95% confidence", "90% confidence"),
  col = c("darkorange", "steelblue", "gray40"),
  lty = c(3, 1, 2), lwd = 2, bty = "n"
)
```

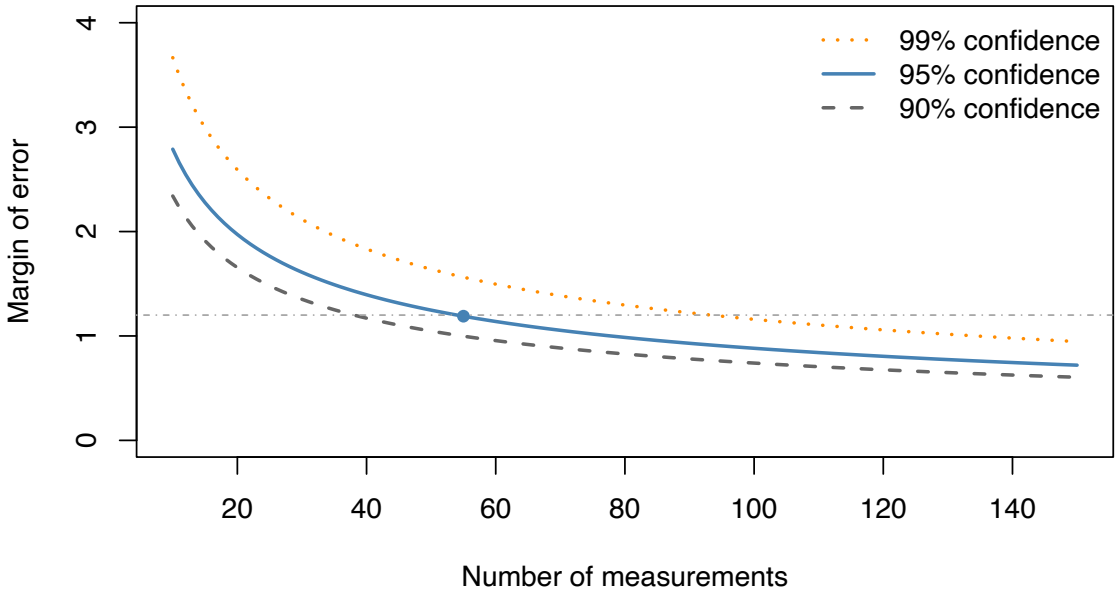


Figure 11.1: More observations reduce the margin of error. A higher confidence level requires a wider interval at every sample size. The horizontal line marks the laboratory's target margin of 1.2 units.

There are two ways to get a narrower interval in Figure 11.1. We can collect more data, or we can accept a lower confidence level. The first improves precision while preserving coverage. The second changes the coverage we are asking the procedure to achieve.

11.3 A normal mean when the variance is unknown

Usually we do not know σ. We estimate it from the sample, which introduces another source of variation. Chapter 10 showed that the t distribution accounts for this variation.

> **Notation: the estimated standard error and t cutoff**
>
> $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$ is the sample variance, and $S = \sqrt{S^2}$ is the sample standard deviation. The estimated standard error of the sample mean is $S/\sqrt{n}$.

$t_p(m)$ is the quantile with probability p to its left in a t distribution with m degrees of freedom. The notation $t_{1-\alpha/2}(n-1)$ therefore specifies both a cumulative probability and the degrees of freedom.
Lowercase $\overline{x}$ and s denote the observed sample mean and sample standard deviation.

For an independent normal sample with $n \geq 2$,

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

The central probability statement is

$$P\left\{-t_{1-\alpha/2}(n-1) \leq \frac{\overline{X} - \mu}{S/\sqrt{n}} \leq t_{1-\alpha/2}(n-1)\right\} = 1 - \alpha.$$

Multiplying by $S/\sqrt{n}$ and isolating μ gives

$$\boxed{\overline{X} \pm t_{1-\alpha/2}(n-1)\frac{S}{\sqrt{n}}.}$$

The t cutoff is larger than the corresponding normal cutoff. It compensates for estimating the standard deviation. Simply replacing σ by S while keeping the normal cutoff does not give an exact confidence interval.

Unlike the known-variance interval, the t interval has a random width. Both its center and its estimated standard error depend on the sample.

11.3.1 Worked example: comparing two fan settings within each computer

An engineer tests two fan settings on each of ten computers under the same workload. For each computer, she subtracts the temperature under the new setting from the temperature under the original setting. A positive difference means the new setting ran cooler.

The observed temperature reductions, in degrees Celsius, are

$$2.2,\ 1.4,\ 3.1,\ 0.8,\ 2.5,\ 1.7,\ 2.9,\ 1.0,\ 2.3,\ 1.6.$$

Assume the differences are independent across computers and normally distributed. The two temperatures measured on the same computer can be dependent. We keep those measurements together and analyze one difference per computer.

> **Notation: paired differences**
>
> D_i is the temperature reduction for computer i. The subscript identifies the computer. $\mu_D = E[D_i]$ is the population mean reduction. The subscript D tells us that the mean concerns differences rather than either temperature alone.
> $\overline{D}$ and S_D are the sample mean and sample standard deviation of the differences. Their observed values are $\overline{d}$ and s_D.

Here

$$\overline{d} = 1.95, \qquad s_D \approx 0.7764, \qquad \frac{s_D}{\sqrt{10}} \approx 0.2455.$$

There are ten independent differences, so the t distribution has nine degrees of freedom. Since $t_{0.975}(9) \approx 2.2622$, the 95% confidence interval for the mean reduction is

$$1.95 \pm 2.2622(0.2455) \approx [1.395,\ 2.505].$$

```
reductions <- c(2.2, 1.4, 3.1, 0.8, 2.5, 1.7, 2.9, 1.0, 2.3, 1.6)
estimated_se <- sd(reductions) / sqrt(10)
margin <- qt(0.975, df = 9) * estimated_se

mean(reductions) + c(-1, 1) * margin
```

```
[1] 1.394606 2.505394
```

The interval estimates the average reduction across the population represented by these computers. It does not say that 95% of individual computers have reductions between these endpoints. An interval for an individual outcome is a prediction interval, like the one developed in Chapter 10.

11.4 What 95% confidence means

Suppose we repeatedly collect independent samples under the same model and use the same 95% confidence interval formula every time. In the long run, about 95% of those intervals will contain the fixed population parameter.

This does not mean exactly 95 of every 100 intervals must succeed. The number that cover is itself random. It also does not mean that an interval contains 95% of the population observations.

Once we have calculated an interval from observed data, both endpoints are fixed numbers. In the frequentist model, the parameter is fixed too. The interval either contains it or it does not. We usually do not know which, but that lack of knowledge does not turn the fixed parameter into a random variable in this model.

Thus the phrase "we are 95% confident" refers to how the interval was constructed. It is shorthand for the procedure's coverage, rather than a posterior probability that this particular interval contains the parameter.

11.4.1 A coverage experiment with a planning consequence

The sample-size calculation above raises a useful question. Does a larger sample make a 95% interval more likely to cover the mean, or does it make the interval more precise?

We can investigate by simulating the laboratory's measurements with a true mean of 30 and standard deviation 4.5. The value 30 is chosen only so that we can check coverage. It is not supplied to the interval formula.

```
set.seed(11132)
sample_sizes <- c(15, 55, 150)
repetitions <- 10000
coverage <- numeric(3)
interval_width <- numeric(3)

for (j in seq_along(sample_sizes)) {
  n <- sample_sizes[j]
  sample_means <- replicate(repetitions, mean(rnorm(n, mean = 30, sd = 4.5)))
  margin <- qnorm(0.975) * 4.5 / sqrt(n)
  lower <- sample_means - margin
  upper <- sample_means + margin
  coverage[j] <- mean(lower <= 30 & 30 <= upper)
  interval_width[j] <- 2 * margin
}

data.frame(
  sample_size = sample_sizes,
  estimated_coverage = round(coverage, 3),
  interval_width = round(interval_width, 3)
)
```

```
  sample_size estimated_coverage interval_width
1          15              0.948          4.555
```

```
2          55              0.950          2.379
3         150              0.954          1.440
```

The coverage estimates should all be close to 0.95. The widths decrease as the sample size grows. The larger samples give more precise intervals while retaining the same confidence level.

Exact coverage depends on the model assumptions. The normal and t calculations above assume independent normal observations. With a nonnormal population, a mean interval may still be a useful large-sample approximation, but its exact coverage no longer follows from these formulas.

11.5 Pivotal quantities: a general way to build intervals

The normal and t constructions used the same idea. We combined the data and the unknown parameter into a quantity with a known distribution. We then rearranged a probability statement about that quantity.

> **Notation: a pivotal quantity**
>
> $Q(\mathbf{X}, \theta)$ denotes a function of the random sample and the parameter we want to estimate. It is called a pivotal quantity, or pivot, if its sampling distribution does not depend on any unknown parameter.
> The symbols c_1 and c_2 denote fixed cutoffs chosen from that known distribution. The subscripts distinguish the lower and upper cutoffs.

A pivot can contain an unknown parameter even though its distribution does not depend on that parameter. For example, $(\overline{X} - \mu)/(\sigma/\sqrt{n})$ contains μ, but it has a standard normal distribution for every value of μ under the model.

To construct an interval, choose cutoffs satisfying

$$P\{c_1 \le Q(\mathbf{X}, \theta) \le c_2\} = 1 - \alpha.$$

Then solve the inequalities for θ. When the resulting set is an interval, its endpoints give a confidence interval.

An **equal-tail** choice leaves probability $\alpha/2$ below c_1 and $\alpha/2$ above c_2. This is a convenient convention. It is not the only possible choice, and it does not always produce the shortest interval.

11.5.1 A confidence interval for a normal population variance

For an independent normal sample with unknown mean and variance, Chapter 9 gives the pivot

$$Q = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

> **Notation: chi-square cutoffs**
>
> $q_p(m)$ denotes the chi-square quantile with m degrees of freedom and probability p to its left, following the convention in Chapter 8.

An equal-tail probability statement is

$$P\left\{q_{\alpha/2}(n-1) \le \frac{(n-1)S^2}{\sigma^2} \le q_{1-\alpha/2}(n-1)\right\} = 1-\alpha.$$

All the quantities in the inequalities are positive. Taking reciprocals reverses the inequalities. Multiplying by $(n-1)S^2$ then gives

$$\boxed{\left[\frac{(n-1)S^2}{q_{1-\alpha/2}(n-1)}, \frac{(n-1)S^2}{q_{\alpha/2}(n-1)}\right].}$$

Notice the order of the denominators. The larger quantile belongs in the lower endpoint because dividing by a larger positive number gives a smaller result.

The interval need not be symmetric around S^2. The chi-square distribution is not symmetric, and the reciprocal transformation changes the spacing further. To obtain a confidence interval for the standard deviation σ, take the positive square root of each endpoint. This preserves coverage because squaring and taking positive square roots preserve order for positive numbers.

11.5.2 Worked example: estimating shared measurement noise

Two independent calibration sessions measure different reference objects. The objects have different unknown mean readings, but suppose the instrument has the same unknown noise variance in both sessions. Assume independent normal readings within and between sessions.

The first session has seven readings and a sample standard deviation of 0.8 millimeters. The second has ten readings and a sample standard deviation of 1.1 millimeters. We want a 95% confidence interval for the common noise standard deviation.

We should not calculate deviations from one overall mean. That would mix differences between the reference objects with the instrument's measurement noise. Instead, we measure variation around each session's own mean.

> **Notation: pooling variation from separate groups**
>
> S_A^2 and S_B^2 are the sample variances in sessions A and B. Their observed values are s_A^2 and s_B^2.
> W denotes the sum of the squared deviations calculated separately within the two sessions. Its observed value is w.
> m denotes the total degrees of freedom after estimating one mean in each session. Here $m = (7-1) + (10-1) = 15$.

Each session provides a chi-square quantity:

$$\frac{6S_A^2}{\sigma^2} \sim \chi_6^2, \qquad \frac{9S_B^2}{\sigma^2} \sim \chi_9^2.$$

These quantities are independent because the sessions are independent. By the addition rule from Chapter 8,

$$\frac{W}{\sigma^2} = \frac{6S_A^2 + 9S_B^2}{\sigma^2} \sim \chi_{15}^2.$$

The observed squared-deviation sum is

$$w = 6(0.8)^2 + 9(1.1)^2 = 14.73.$$

Using $q_{0.025}(15) \approx 6.2621$ and $q_{0.975}(15) \approx 27.4884$, the confidence interval for the common variance is

$$\left[\frac{14.73}{27.4884}, \frac{14.73}{6.2621}\right] \approx [0.5359,\ 2.3522]$$

in squared millimeters. Taking square roots gives the requested interval for the common standard deviation:

$$\boxed{[0.732,\ 1.534] \text{ millimeters.}}$$

```r
within_sum <- 6 * 0.8^2 + 9 * 1.1^2
variance_interval <- within_sum / qchisq(c(0.975, 0.025), df = 15)
```

```r
sqrt(variance_interval)
```

```
[1] 0.7320263 1.5336987
```

The total degrees of freedom are 15 because two means were estimated from seventeen readings. This shared interval relies on the assumption of a common variance. Independence alone would not justify pooling the sessions in this way.

11.6 Bayesian credible intervals

Bayesian inference gives a probability distribution for the unknown parameter after combining a prior with the observed data. An interval can then be chosen to contain a specified amount of posterior probability.

> **Notation: posterior probability and credible endpoints**
>
> $\pi(\theta \mid \mathbf{x})$ denotes the posterior density of θ given the observed data $\mathbf{x}$. The vertical bar means "given," and π here names a density rather than the mathematical constant. l_B and u_B denote numerical endpoints of a Bayesian credible interval. The subscript B distinguishes these endpoints from the confidence interval endpoints discussed earlier. The expression $\int_{l_B}^{u_B} \pi(\theta \mid \mathbf{x})\, d\theta$ is the area under the posterior density between the two endpoints. It is a posterior probability.

A $100(1-\alpha)\%$ credible interval satisfies

$$P(l_B \le \theta \le u_B \mid \mathbf{x}) = \int_{l_B}^{u_B} \pi(\theta \mid \mathbf{x})\, d\theta = 1 - \alpha.$$

For a continuous posterior, an equal-tail credible interval uses the posterior $\alpha/2$ and $1-\alpha/2$ quantiles. It leaves the same posterior probability outside each end.

Here it is appropriate to say that the parameter has posterior probability $1-\alpha$ of lying in the observed interval, given the data and the specified model and prior. This interpretation follows from the Bayesian view of probability.

11.6.1 Worked example: an interval for a seed germination probability

A greenhouse tests 24 seeds under a new storage protocol, and 18 germinate. Model the outcomes as independent Bernoulli trials with a common germination probability. Before seeing these observations, suppose the greenhouse uses a Beta$(2, 2)$ prior.

> **Notation: the germination model**
>
> p is the probability that a seed germinates.
> K is the number that germinate among the 24 seeds. The notation $K \mid p \sim \text{Binomial}(24, p)$ describes its distribution when p is given.
> Beta(a, b) is a beta distribution with positive shape parameters a and b, as in Chapter 3. These shape parameters determine the density on the interval from 0 to 1.

The beta-binomial update from Chapter 3 gives

$$p \mid K = 18 \sim \text{Beta}(2 + 18,\ 2 + 6) = \text{Beta}(20, 8).$$

The equal-tail 90% credible interval uses the 0.05 and 0.95 posterior quantiles:

```
qbeta(c(0.05, 0.95), shape1 = 20, shape2 = 8)
```

```
[1] 0.567698 0.843181
```

The resulting interval is approximately

$$[0.568,\ 0.843].$$

Under this model and prior, the posterior probability that the germination probability lies between these endpoints is 0.90. The interval describes uncertainty about a probability, rather than a range containing 90% of future seed outcomes.

Suppose the greenhouse is also interested in whether the germination probability exceeds 0.60. The posterior can answer that question directly:

```
pbeta(0.60, shape1 = 20, shape2 = 8, lower.tail = FALSE)
```

```
[1] 0.9047121
```

This posterior probability is about 0.905. It comes from the posterior area above 0.60. It is a different calculation from asking for an interval with 90% posterior probability.

A different prior can give a different credible interval from the same observations. The prior is part of the probability model, so the interpretation should acknowledge it. Also, a 90% credible interval does not automatically have 90% frequentist coverage for every fixed value of the parameter.

11.7 Choosing and explaining an interval

Before calculating endpoints, identify the quantity you want to estimate and the sampling assumptions you are making. A population mean, a population standard deviation, and a future observation require different intervals.

Goal and assumptions	Interval construction
Mean of a normal population, known standard deviation	Sample mean plus or minus a normal cutoff times the known standard error
Mean of a normal population, unknown standard deviation	Sample mean plus or minus a t cutoff times the estimated standard error
Variance of a normal population	Invert a chi-square probability statement
Parameter with a specified continuous posterior	Choose endpoints containing the desired posterior probability

For a confidence interval, explain the coverage of the procedure. For a credible interval, explain the posterior probability conditional on the observed data. In either case, include the units, the confidence or credibility level, and the assumptions needed to support the calculation.

12 Unbiased Estimators

An estimator is a rule for turning a sample into a guess about an unknown population quantity. If we collect a different sample, the rule usually gives a different answer. To judge the rule, we need to consider how it behaves across possible samples.

One appealing property is that its answers average to the correct value. An estimator with this property is called **unbiased**. Some samples produce estimates that are too large, and others produce estimates that are too small, but the signed errors balance in expectation.

Unbiasedness is useful, but it does not tell the whole story. An estimator can be correct on average while making large errors in both directions. This chapter develops two ways to describe an estimator: its bias, which measures systematic displacement, and its mean squared error, which measures the overall size of its errors.

12.1 What unbiasedness means

> **i** Notation: an estimator and its sampling expectation
>
> θ is the fixed, unknown parameter to be estimated. The bold symbol $\mathbf{X} = (X_1, \dots, X_n)$ represents the random sample.
> $\delta(\mathbf{X})$ is an estimator, meaning a rule applied to the sample. The Greek letter δ, pronounced "delta," names the rule. We may also write an estimator as $\widehat{\theta}$. The hat marks a quantity used to estimate the parameter below it.
> $E_\theta[\delta(\mathbf{X})]$ is the mean of the estimator's sampling distribution when the true parameter is θ. The subscript reminds us that the distribution of the data depends on θ.

An estimator is unbiased for θ if

$$\boxed{E_\theta[\delta(\mathbf{X})] = \theta}$$

for every allowed value of θ. We assume the expectation exists.

The phrase "for every allowed value" matters. An estimator that happens to be centered correctly at one parameter value is not necessarily unbiased throughout the model. For instance,

the rule that always returns zero has the correct expectation when $\theta = 0$, but it is not an unbiased estimator of an unrestricted population mean.

Unbiasedness concerns the rule before the data are observed. After seeing the data, we obtain one numerical estimate. We cannot tell whether that estimate is close to the truth just by knowing that the rule is unbiased.

It also matters which quantity we are estimating. A statistic can be unbiased for a variance and biased for a standard deviation. The target must be stated before an unbiasedness claim has a clear meaning.

12.2 Bias measures systematic displacement

> **Notation: bias**
>
> $\text{Bias}_\theta(\hat{\theta})$ denotes the bias of an estimator when the parameter is θ:
>
> $$\text{Bias}_\theta(\hat{\theta}) = E_\theta[\hat{\theta}] - \theta.$$
>
> The expression $\hat{\theta} - \theta$ is the estimation error in one sample. Bias is the expectation of that signed error across samples.

A positive bias means the estimator is too large on average. A negative bias means it is too small on average. Zero bias means the positive and negative errors balance in expectation.

Bias and estimation error are different. An unbiased estimator usually has a nonzero error in any particular sample. A biased estimator can sometimes give an estimate very close to the true parameter.

Nor does zero bias guarantee equal probabilities of overestimation and underestimation. Expectations account for the sizes of errors as well as their probabilities. Many small negative errors can balance a few large positive errors.

12.2.1 Two familiar results, with their assumptions

We have already encountered two important unbiased estimators. It helps to separate what follows from expectations alone from what requires a normal population.

> **Notation: sample mean and sample variance**
>
> $\mu = E[X_i]$ is the common population mean, and $\sigma^2 = \text{Var}(X_i)$ is the common population variance.

$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is the sample mean.
$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$ is the sample variance for $n \geq 2$.

If all the observations have the same finite mean μ, linearity of expectation gives

$$E[\overline{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{n\mu}{n} = \mu.$$

Thus the sample mean is unbiased. Independence is not needed for this expectation calculation, although it is needed for many of the usual formulas for the sample mean's variance.

For independent observations from the same population with finite variance σ^2, the sample variance is also unbiased:

$$E[S^2] = \sigma^2.$$

Here is why the divisor is $n-1$. The squared-deviation identity from Chapter 9 gives

$$\sum_{i=1}^{n}(X_i - \overline{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\overline{X} - \mu)^2.$$

The expected value of the first term on the right is $n\sigma^2$. Since $\overline{X}$ has mean μ and variance σ^2/n, the expected value of the second term is

$$nE[(\overline{X} - \mu)^2] = n\frac{\sigma^2}{n} = \sigma^2.$$

The expected squared-deviation sum is therefore $(n-1)\sigma^2$. Dividing by $n-1$ gives the desired result.

These unbiasedness results do not require a normal distribution. Normality was needed in Chapter 9 for the exact chi-square distribution of the scaled sample variance and for independence of the sample mean and variance. Those are stronger conclusions.

Taking square roots does not carry unbiasedness from S^2 to S. In general, the expectation of a nonlinear function is not the same as that function applied to the expectation. We will see this issue in a worked example below.

12.3 Unbiased estimators can have very different variability

If two estimators are both unbiased, knowing their biases does not help us choose between them. We also need to examine how widely their estimates vary.

12.3.1 Worked example: combining instruments with different precision

Two instruments measure the same temperature. Both are correctly calibrated, but the first is more precise. Suppose the readings are independent, each has expectation equal to the true temperature, and their variances are 0.16 and 0.64 squared degrees Celsius.

How should we combine the readings?

> **Notation: a weighted estimate**
>
> A and B are the readings from the first and second instruments. The unknown true temperature is θ, so $E[A] = E[B] = \theta$.
> w is a fixed weight between 0 and 1. The estimator
>
> $$\hat{\theta}_w = wA + (1 - w)B$$
>
> places weight w on the first reading and weight $1 - w$ on the second. The subscript w labels which weighting rule we use.

Because the weights sum to 1,

$$E[\hat{\theta}_w] = wE[A] + (1 - w)E[B] = w\theta + (1 - w)\theta = \theta.$$

Every fixed choice of w gives an unbiased estimator. Unbiasedness alone leaves us with many choices.

Independence gives the variance

$$\mathrm{Var}(\hat{\theta}_w) = 0.16w^2 + 0.64(1 - w)^2.$$

Using equal weights, $w = 0.5$, gives

$$\mathrm{Var}(\hat{\theta}_{0.5}) = 0.16(0.5)^2 + 0.64(0.5)^2 = 0.20.$$

To find the best weight within this family, rewrite the variance as

$$0.16w^2 + 0.64(1 - w)^2 = 0.80(w - 0.80)^2 + 0.128.$$

The squared term cannot be negative. It is zero at $w = 0.80$, so this choice gives the smallest variance, 0.128. The more precise instrument receives more weight.

If the observed readings are 20.4 and 19.8 degrees, the equal-weight estimate is 20.1 degrees, while the lower-variance estimate is

$$0.8(20.4) + 0.2(19.8) = 20.28 \text{ degrees.}$$

We cannot know which numerical answer is closer to the true temperature from these readings alone. What we can say is that the 0.8 weighting rule has smaller sampling variance under the assumptions. It is more precise across repeated measurements.

12.4 Transforming an unbiased estimator can introduce bias

Linearity of expectation makes sums and fixed multiples relatively easy to handle. Squaring, taking a square root, or taking a reciprocal requires more care.

For a random variable with finite variance,

$$E[\overline{X}^2] = \operatorname{Var}(\overline{X}) + (E[\overline{X}])^2.$$

Even if $\overline{X}$ is unbiased for a mean, its square is generally too large on average as an estimator of the squared mean. The extra term is the variance of $\overline{X}$.

12.4.1 Worked example: estimating the expected product of two event counts

A monitoring system records the number of requests arriving in each one-minute period. Model counts from separate periods as independent Poisson random variables with the same unknown mean rate.

The quantity of interest is the expected product of the counts in two independent future periods. Independence makes this expected product equal to the square of the mean rate. We want to estimate that squared rate from past observations.

> **i** Notation: a squared rate and estimates of it
>
> $X_i \sim \text{Poisson}(\lambda)$ means that the count in period i has a Poisson distribution with mean and variance both equal to λ.
> $\psi = \lambda^2$ names the quantity we want to estimate. The Greek letter ψ is pronounced "psi." Using a different symbol helps distinguish the squared rate from the rate itself.
> $\widehat{\psi}_0$ and $\widehat{\psi}_1$ are two estimators of ψ. The hats identify estimators, and the subscripts 0

and 1 label the two rules.
$K = \sum_{i=1}^{n} X_i$ is the total count over the n observed periods, so $\overline{X} = K/n$.

The sample mean is unbiased for λ and has variance λ/n. A natural first attempt is to square it:

$$\widehat{\psi}_0 = \overline{X}^2.$$

Its expectation is

$$E_\lambda[\widehat{\psi}_0] = \mathrm{Var}_\lambda(\overline{X}) + (E_\lambda[\overline{X}])^2 = \frac{\lambda}{n} + \lambda^2.$$

The subscript λ plays the same role that the subscript θ did earlier. It identifies the parameter governing the sampling distribution. The bias for estimating $\psi = \lambda^2$ is λ/n.

We cannot subtract λ/n from our estimate because λ is unknown. However, $\overline{X}/n$ has expectation λ/n. Subtracting that statistic removes the bias:

$$\widehat{\psi}_1 = \overline{X}^2 - \frac{\overline{X}}{n}.$$

Indeed,

$$E_\lambda[\widehat{\psi}_1] = \left(\lambda^2 + \frac{\lambda}{n}\right) - \frac{\lambda}{n} = \lambda^2.$$

This establishes unbiasedness for every allowed rate, rather than for just one numerical choice.

In terms of the total count, the corrected estimator is

$$\widehat{\psi}_1 = \frac{K^2 - K}{n^2} = \frac{K(K-1)}{n^2}.$$

Because K is a nonnegative integer, this estimate is nonnegative. When the total count is zero or one, it equals zero.

Suppose the five observed counts are $0, 1, 2, 0, 1$. Then $K = 4$ and $\overline{x} = 0.8$. The uncorrected estimate of the squared rate is $0.8^2 = 0.64$, while the unbiased estimate is

$$0.8^2 - \frac{0.8}{5} = 0.48.$$

```r
counts <- c(0, 1, 2, 0, 1)
estimated_rate <- mean(counts)

estimated_rate^2
```

```
[1] 0.64
```

```r
estimated_rate^2 - estimated_rate / 5
```

```
[1] 0.48
```

Removing bias does not yet establish that the corrected rule has smaller overall estimation error. That requires a measure that accounts for variability as well as bias.

12.5 Mean squared error

An estimator can make errors in either direction. If we average the signed errors, positive and negative values can cancel. Squaring the errors makes them nonnegative and gives greater weight to large mistakes.

> **Notation: mean squared error**
>
> $\mathrm{MSE}_\theta(\hat{\theta})$ denotes the mean squared error of an estimator:
>
> $$\mathrm{MSE}_\theta(\hat{\theta}) = E_\theta[(\hat{\theta} - \theta)^2].$$
>
> The abbreviation MSE stands for mean squared error. Its units are the squared units of the parameter. For example, estimating a length in centimeters produces an MSE in squared centimeters.
> $\mathrm{Var}_\theta(\hat{\theta})$ is the sampling variance of the estimator. It measures spread around the estimator's own expectation, which need not equal θ.

MSE is an average over possible samples at a fixed true parameter value. It is not the squared error of just the observed estimate. Since the true parameter is unknown, we usually study MSE as a function of that parameter.

This connects to the squared error loss in Chapter 4. The loss for an estimate is its squared error. Here we average that loss over repeated samples while holding the parameter fixed. In a Bayesian posterior expected loss calculation, we instead hold the observed data fixed and average over the posterior distribution of the parameter.

12.5.1 Separating variance and squared bias

The key identity is

$$\boxed{\mathrm{MSE}_\theta(\hat{\theta}) = \mathrm{Var}_\theta(\hat{\theta}) + \left[\mathrm{Bias}_\theta(\hat{\theta})\right]^2.}$$

To see where it comes from, add and subtract the estimator's expectation inside its error:

$$\hat{\theta} - \theta = \left(\hat{\theta} - E_\theta[\hat{\theta}]\right) + \left(E_\theta[\hat{\theta}] - \theta\right).$$

The first term describes random variation around the estimator's own center. The second is its bias. At a fixed parameter value, that second term is a constant.

Squaring and taking expectations produces three terms:

$$\begin{aligned} E_\theta[(\hat{\theta} - \theta)^2] = E_\theta\Big[\left(\hat{\theta} - E_\theta[\hat{\theta}]\right)^2\Big] \\ + 2\left(E_\theta[\hat{\theta}] - \theta\right) E_\theta\left[\hat{\theta} - E_\theta[\hat{\theta}]\right] \\ + \left(E_\theta[\hat{\theta}] - \theta\right)^2. \end{aligned}$$

The first term is the variance. The middle term is zero because a random variable minus its own expectation has expectation zero. The last term is the squared bias. This proves the identity whenever the estimator has a finite second moment.

For an unbiased estimator, MSE equals variance. In the instrument example, the two weighting rules therefore have MSEs 0.20 and 0.128.

For a biased estimator, MSE is larger than its own variance by the amount of its squared bias. That does not mean it has larger MSE than every unbiased estimator. A sufficiently large reduction in variance can more than compensate for a nonzero bias.

12.6 A biased estimator can have smaller MSE

12.6.1 Worked example: stabilizing an estimate from eight trials

A small pilot study tests a new automatic door mechanism eight times under the same conditions. Each trial either completes successfully or fails. Suppose the outcomes are independent with a common success probability.

With only eight trials, the sample proportion can vary substantially. We will compare it with an estimator that pulls the result toward one-half.

Notation: the sample proportion and a smoothed estimate

p is the unknown success probability, and $K \sim \text{Binomial}(8, p)$ is the number of successes in eight trials. For this distribution, $E_p[K] = 8p$ and $\text{Var}_p(K) = 8p(1-p)$.
$\hat{p} = K/8$ is the sample proportion. Its hat identifies it as an estimator of p.
$\tilde{p} = (K+1)/10$ is a second estimator. The tilde, the small wavy mark over p, distinguishes this smoothed estimator from the sample proportion. It does not imply a general statistical property.

The sample proportion is unbiased:

$$E_p[\hat{p}] = p, \qquad \text{Var}_p(\hat{p}) = \frac{p(1-p)}{8}.$$

Its MSE is therefore

$$\text{MSE}_p(\hat{p}) = \frac{p(1-p)}{8}.$$

The smoothed estimate can be written as

$$\tilde{p} = 0.8\hat{p} + 0.2\left(\frac{1}{2}\right).$$

It places 80% of its weight on the sample proportion and 20% on the fixed value one-half. For example, eight successes give a sample proportion of 1 but a smoothed estimate of 0.9. Zero successes give 0 and 0.1, respectively.

The expectation of the smoothed estimator is

$$E_p[\tilde{p}] = \frac{8p+1}{10}.$$

Its bias is

$$\text{Bias}_p(\tilde{p}) = \frac{8p+1}{10} - p = \frac{1-2p}{10}.$$

When $p < 1/2$, this bias is positive. When $p > 1/2$, it is negative. In both cases, the estimator's expectation is pulled toward one-half. At $p = 1/2$ the bias is zero, but the estimator is not unbiased across the whole model.

Adding a constant does not change variance, while dividing by 10 divides variance by 100. Thus

$$\mathrm{Var}_p(\tilde{p}) = \frac{8p(1-p)}{100}.$$

This is 64% of the sample proportion's variance. Adding the squared bias to the variance gives

$$\mathrm{MSE}_p(\tilde{p}) = \frac{8p(1-p) + (1-2p)^2}{100}.$$

Now compare the MSEs at two possible parameter values:

True success probability	MSE of $\hat{p}$	Bias of $\tilde{p}$	MSE of $\tilde{p}$	Smaller MSE
$p = 0.50$	0.03125	0	0.02000	Smoothed estimator
$p = 0.05$	0.00594	0.09	0.01190	Sample proportion

Near one-half, the reduction in variance outweighs the squared bias. Near zero or one, pulling toward one-half can introduce too much bias. Neither estimator has smaller MSE at every parameter value.

```
probabilities <- seq(0, 1, length.out = 501)
mse_proportion <- probabilities * (1 - probabilities) / 8
variance_smoothed <- 8 * probabilities * (1 - probabilities) / 100
bias_smoothed <- (1 - 2 * probabilities) / 10
mse_smoothed <- variance_smoothed + bias_smoothed^2

plot(
  probabilities, mse_proportion,
  type = "l", col = "steelblue", lwd = 2,
  ylim = c(0, 0.04), xlab = "True success probability",
  ylab = "Mean squared error"
)
lines(probabilities, mse_smoothed, col = "darkorange", lwd = 2, lty = 2)
legend(
  "topright", legend = c("Sample proportion", "Smoothed estimate"),
  col = c("steelblue", "darkorange"), lwd = 2, lty = c(1, 2), bty = "n"
)
```

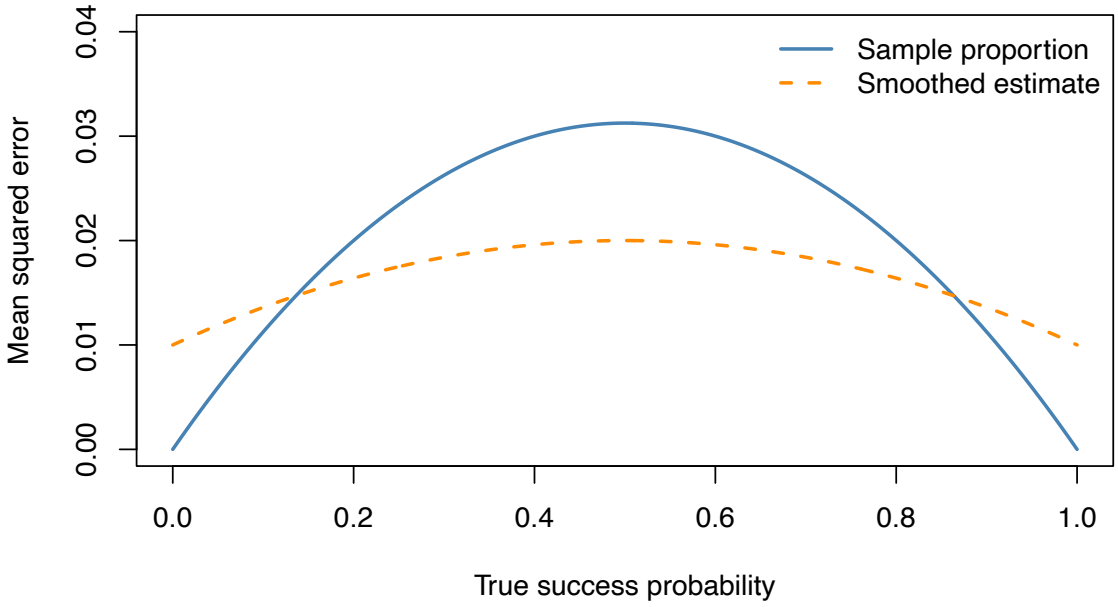


Figure 12.1: Smoothing reduces variance at every success probability, but it also introduces bias. The smoothed estimate has smaller MSE in the middle of the parameter range and larger MSE near its ends.

Figure 12.1 compares procedures across possible true values of p. The horizontal axis is not an observed estimate. In practice, we do not know where the true parameter lies on that axis. The graph tells us how the relative performance depends on that unknown value.

12.6.2 Checking the MSE calculation by simulation

We can also generate many pilot studies at a chosen success probability and average the squared errors. The simulation uses a known value of p so that the errors can be calculated.

```
set.seed(12132)
probabilities_to_check <- c(0.05, 0.50)
simulated_mse_proportion <- numeric(2)
simulated_mse_smoothed <- numeric(2)

for (j in seq_along(probabilities_to_check)) {
```

```
  true_probability <- probabilities_to_check[j]
  successes <- rbinom(100000, size = 8, prob = true_probability)
  proportion_estimates <- successes / 8
  smoothed_estimates <- (successes + 1) / 10

  simulated_mse_proportion[j] <- mean((proportion_estimates - true_probability)^2)
  simulated_mse_smoothed[j] <- mean((smoothed_estimates - true_probability)^2)
}

data.frame(
  true_probability = probabilities_to_check,
  mse_proportion = round(simulated_mse_proportion, 5),
  mse_smoothed = round(simulated_mse_smoothed, 5)
)
```

```
  true_probability mse_proportion mse_smoothed
1             0.05        0.00594      0.01185
2             0.50        0.03147      0.02014
```

The simulated values should be close to the exact MSEs in the table. Small differences are ordinary simulation variation. Increasing the number of repetitions makes that variation smaller, but does not change either estimator or the size of each pilot study.

12.7 How to evaluate an estimator

When studying an estimator, first state the target. Then find its expectation under the sampling model. Subtracting the target gives the bias. Next find the variance, and combine variance with squared bias to obtain MSE.

Quantity	What it measures	Question it answers
Bias	Displacement of the estimator's expectation from the target	Is the rule systematically high or low?
Variance	Spread around the estimator's own expectation	How much does the estimate change across samples?
Mean squared error	Average squared distance from the target	How large are the errors overall under squared error loss?

Unbiasedness is one useful property of a procedure. It does not guarantee that a particular estimate is accurate, and it does not settle a comparison between procedures. To understand how well an estimator works, we need to consider both where its sampling distribution is centered and how widely it spreads.